

Sukhrob Ilyosbekov

Senior Software Engineer · AI/ML & Computer Vision

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SUMMARY

Machine-learning engineer with a published computer-vision research papers and 9+ years building production software. I take deep-learning and LLM work past the prototype stage: imaging models pulled into hospital workflows, an AI freight dispatcher, and multi-model pipelines running in live SaaS. The full-stack background is what gets the models deployed and keeps them running once they're out.

TECHNICAL SKILLS

AI/ML: PyTorch, TensorFlow, EfficientNet, YOLO, GradCAM++, OpenCV, computer vision, LLM/agent systems, MCP, Claude/GPT/DeepSeek APIs

Languages: Python, C#, C++, TypeScript, JavaScript, Kotlin

ML Tooling & Data: FastAPI, OHIF/DICOM, PostgreSQL, MongoDB, Redis, Prisma, NumPy/pandas

Backend: ASP.NET Core, EF Core, SignalR, Node.js, Bun, NestJS, ElysiaJS

Frontend: React, Next.js, Angular, Blazor, Tailwind CSS

Cloud & DevOps: AWS (S3, Lambda, Cognito, Amplify), Azure, Docker, Kubernetes, GitHub Actions

EDUCATION

Northeastern University, M.S. in Computer Science, Boston, MA

Jan 2024 – May 2026

Suffolk University, B.S. in Computer Science, Boston, MA

Sep 2021 – May 2023

PROFESSIONAL EXPERIENCE

Senior AI/ML Engineer | **EmTech Care Labs** | Portland, ME

Jan 2025 – Jun 2025

- Built the machine-learning features for a HIPAA-compliant Alzheimer's care platform, including a risk model that flags patients showing early signs of decline from daily activity and care-log data so caregivers can step in sooner.
- Added an NLP pipeline that reads free-text clinical notes and pulls out structured details (symptoms, medications, care events), cutting the manual charting load for the care team.
- Stood up the ML infrastructure under HIPAA constraints, with de-identified training data, model serving behind the API, and monitoring. The platform cleared an outside HIPAA readiness review with no critical findings.

Senior Machine Learning Engineer | **AllFactors** | Santa Clara, CA

July 2022 – Dec 2023

- Built property-valuation and market-forecasting models on real-estate data that powered the product's analytics, turning raw pricing and inventory feeds into predicted values and trend signals.
- Designed the real-time feature pipeline (SignalR + SQL Server change tracking) that fed live market data to the models and pushed updated predictions to hundreds of concurrent users with sub-second latency.
- Productionized the models into the customer-facing dashboards and added automated tests around the data and prediction flows.

Software Engineer | **Smart Meal Service** | Russia

Sep 2020 – Oct 2021

- Built the software for a self-service food kiosk and robotic cashier in C# / .NET, integrating with payment terminals and hardware over low-level protocols, plus the customer ordering app and an admin dashboard.
- Added automated order routing that noticeably cut kitchen wait times, and a hardware abstraction layer so one codebase ran across three machine models.

Game Developer | **Pentalight Technology** | Malaysia

Mar 2020 – Feb 2021

- Built the multiplayer networking for a VR smart-city simulation in Unity, keeping many users in sync in real time with low latency, plus spatial audio, hand-tracking, and a desktop spectator client.

Earlier: Freelance Developer at Freelancer.com (2019 to 2020), building Python NLP and .NET applications. Game Developer at EC Dev Team, Uzbekistan (2016 to 2019), led RTS game development in the Clausewitz Engine.

SELECTED AI/ML PROJECTS

MelanomaNet

[GitHub](#) | [Paper](#)

PyTorch, EfficientNet V2, GradCAM++, OpenCV

- Trained a deep-learning model to classify skin lesions across nine diagnostic categories on a standard public dermatology dataset of 25K+ images, reaching roughly 86% accuracy.
- The research contribution is interpretability: the model produces heatmaps of what it looked at, decomposed along the same ABCDE criteria (Asymmetry, Border, Color, Diameter, Evolution) dermatologists already use, so a clinician can follow and trust the reasoning rather than getting an opaque score. Published on arXiv.

Med Image Scanner

<https://github.com/suxrobgm/med-image-scanner>

FastAPI, PyTorch, OpenCV, OHIF Viewer, DICOM, Next.js

- HIPAA-compliant platform that pulls X-ray, CT, and MRI scans directly from hospital imaging systems and runs PyTorch detection models on them, flagging pneumonia on chest X-rays and intracranial hemorrhage on head CTs.
- Renders model predictions as overlays inside a real radiology viewer (OHIF), and handles patient data safely throughout, with on-the-fly de-identification, audit logging, and role-based access.

Bookshelf Scanner

<https://github.com/suxrobgm/bookshelf-scanner>

YOLO, Moondream2 VLM, FastAPI, Angular

- Two-stage computer-vision pipeline: a YOLO segmentation model isolates each individual book spine from a shelf photo, then a vision-language model reads the title and author off each spine.
- Wrapped it in a web app for uploading photos, correcting misreads, and exporting the recognized library to Goodreads or LibraryThing.

LogisticsX

<https://logisticsx.app> | [GitHub](#)

Claude API, MCP, .NET 10, EF Core, SignalR, Angular 21, Kotlin KMP, PostgreSQL

- Built an AI dispatcher that matches freight loads to trucks, checks federal driver hours-of-service rules, and plans multi-stop routes on its own, turning a manual 15-minute decision into a near-instant one. Runs on the Claude API through a custom tool-use agent with its own set of tools.
- Shipped it as a real product: web portals for dispatchers, customers, and brokers, a cross-platform driver app, and a multi-tenant .NET backend integrated with the major freight load boards and tracking providers.

OGStack

<https://ogstack.dev> | [GitHub](#)

Next.js 16, Elysia, Bun, Satori, Cloudflare R2, Stripe

- Built a multi-model AI layer that routes work across Claude, GPT, and DeepSeek to read a web page and match its branding, then generates an on-brand social-share image, including AI-generated backgrounds.
- Productionized it as a SaaS serving cached images from a global CDN; it generated 10K+ images during the beta.